Employee Attrition Prediction Using Machine Learning

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Introduction

Employee attrition is a major challenge for organizations because losing skilled employees can lead to increased hiring costs, reduced productivity, and disruption in business operations. Understanding why employees leave and predicting attrition can help management take proactive actions to improve employee retention.

The objective of this study is to predict employee attrition using machine learning techniques and identify the key factors that influence employees' decisions to leave the company. By understanding these factors, management can develop strategies to reduce turnover and improve employee satisfaction.

Data Source

The dataset used in this project is the **IBM HR Analytics Employee Attrition & Performance** dataset obtained from Kaggle. This is a real-world business dataset containing employee demographic information, job-related characteristics, compensation details, and satisfaction metrics.

The dataset contains **1,470 employee records** and multiple variables such as Age, Monthly Income, Overtime, Job Satisfaction, Work-Life Balance, and Attrition status.

The target variable in this study is **Attrition**, which indicates whether an employee has left the company.

Methodology

The data was first preprocessed by converting categorical variables into factors and removing unnecessary variables such as EmployeeCount, StandardHours, Over18, and EmployeeNumber.

The dataset was then split into **80% training data** and **20% testing data**. The training data was used to build machine learning models, while the testing data was used to evaluate model performance on unseen data.

Two machine learning techniques were applied:

1. **Logistic Regression**
   Logistic Regression was used to estimate the probability of employee attrition and identify statistically significant predictors.
2. **Random Forest**
   Random Forest was used as an ensemble machine learning model to improve prediction and identify variable importance.

The models were evaluated using a confusion matrix and performance metrics such as Accuracy, Sensitivity, and Specificity.

Results

The Logistic Regression model achieved an accuracy of **89.8%**, outperforming the Random Forest model, which achieved an accuracy of **85.7%**.

Logistic Regression showed better balance in identifying employee attrition cases and was selected as the preferred model for this study.

The analysis revealed several significant factors that increase employee attrition, including:

- Overtime
- Frequent Business Travel
- Distance From Home
- Years Since Last Promotion
- Number of Companies Worked
- Being Single

Factors that reduce employee attrition include:

- Higher Age
- Better Job Satisfaction
- Better Environment Satisfaction
- Better Work-Life Balance
- Higher Job Involvement
- More Training Opportunities

Variable importance analysis using Random Forest showed that the most influential variables were:

- Monthly Income
- Age
- Total Working Years
- Job Role
- Overtime
- Distance From Home

Conclusion and Recommendations

Based on the analysis, employee attrition can be predicted with high accuracy using machine learning techniques. Logistic Regression was found to be the most effective model for this study.

The results suggest that management should focus on the following strategies to reduce attrition:

- Reduce employee overtime where possible
- Improve work-life balance initiatives
- Review salary and compensation structures
- Provide more promotion opportunities
- Offer remote or hybrid work options to reduce commuting stress
- Increase employee engagement and satisfaction programs

Implementing these recommendations can help organizations reduce employee turnover, improve employee satisfaction, and lower recruitment and training costs.